Lab2

Theory:

1. Differentiate between one-dimensional and multidimensional arrays.
2. How are array variables declared and initialized in C? Provide examples for both one-dimensional and multidimensional arrays.
3. Explain how to access elements in a one-dimensional and a two-dimensional array.
4. How are string variables declared and initialized in C?
5. Explain the difference between a character array and a string in C.
6. Discuss at least five string handling functions in C (e.g., strlen(), strcpy(), strcat(), strcmp(), strrev()) and their uses.
7. Explain the following sorting algorithms with their working principles:
   a) Bubble Sort
   b) Selection Sort
   c) Insertion Sort
   d) Merge Sort
8. Compare the efficiency of these algorithms in terms of time complexity (best case, average case, and worst case).

Lab questions:

1. Write a C program to declare and initialize a one-dimensional array of integers and print all its elements.
2. Write a C program to find the sum of all elements in a one-dimensional array.
3. Write a C program to find the largest and smallest element in a one-dimensional array.
4. Write a C program to declare a 2D array (matrix) and print its elements in matrix form.
5. Write a C program to find the sum of two 2D arrays (matrices) of the same size.
6. Write a C program to multiply two 2D arrays (matrices) and display the result.
7. Write a C program to find the transpose of a 2D array (matrix).
8. Write a C program to check if a 2D array is symmetric (i.e., matrix equals its transpose).
9. Write a C program to count the number of even and odd numbers in a one-dimensional array.
10. Write a C program to reverse the elements of a one-dimensional array.
11. Write a C program to declare and initialize a string and print its length.
12. Write a C program to concatenate two strings without using the strcat() function.
13. Write a C program to compare two strings without using the strcmp() function.
14. Write a C program to copy one string to another without using the strcpy() function.
15. Write a C program to count the number of vowels and consonants in a string.
16. Write a C program to check if a string is a palindrome.
17. Write a C program to reverse a string without using the strrev() function.
18. Write a C program to count the number of words in a string.
19. Write a C program to convert a string to uppercase without using the strupr() function.
20. Write a C program to find the frequency of a specific character in a string.
21. Write a C program to implement the Bubble Sort algorithm to sort an array of integers.
22. Write a C program to implement the Selection Sort algorithm to sort an array of integers.
23. Write a C program to implement the Insertion Sort algorithm to sort an array of integers.
24. Write a C program to implement the Merge Sort algorithm to sort an array of integers.
25. Write a C program to compare the efficiency of Bubble Sort and Selection Sort by counting the number of swaps and comparisons.
26. Write a C program to sort an array of strings using the Bubble Sort algorithm.
27. Write a C program to sort an array of integers in descending order using Insertion Sort.
28. Write a C program to merge two sorted arrays into a single sorted array.
29. Write a C program to find the time complexity of Merge Sort for different input sizes.
30. Write a C program to sort a 2D array (matrix) row-wise using the Selection Sort algorithm.